

Curriculum Vitae

Sarah Maria Morell

Researcher at University of Bremen, Germany

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ACADEMIC POSITIONS

Researcher (with tenure track to Senior Researcher), 0.75 FTE Oct 2025 – present
University of Bremen, Germany

Postdoctoral Researcher Apr 2025 – Sep 2025
University of Bremen, Germany
Supervisor: Prof. Dr. Nicole Megow

EDUCATION

PhD in Mathematics Sep 2018 – Mar 2025
Technische Universität Berlin, Germany
PhD Thesis: *Flow-based Problem Solving in Combinatorial Optimization*
Supervisor: Prof. Dr. Martin Skutella

Career interruptions following the birth of children
Parental leave Apr – Sep 2021, 0.5 FTE Oct 2021 – Nov 2022
Parental leave Apr 2023 – Apr 2024, 0.6 FTE May – Jul 2024

Master in Mathematics (and Minor in Computer Science) Sep 2015 – Jul 2018
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Master Thesis: *Algorithms for Diversity Maximization*
Supervisor: Prof. Dr. Friedrich Eisenbrand

Bachelor in Mathematics Sep 2009 – Feb 2014
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

Abitur (High School Diploma) – Jul 2008
Gymnasium Eckhorst, Bargteheide, Germany

RESEARCH STAYS

Maastricht University, Netherlands Feb – Mar 2023
Visiting Prof. Dr. Lars Rohwedder

Technical University Munich, Germany Mar – Apr 2020
Visiting Prof. Dr. Stefan Weltge

PUBLICATIONS

- Unsplittable Transshipments*** with S. Degupta and M. Skutella 2026
In: *Proceedings of ICALP 2026*.
- Flow-based Problem Solving in Combinatorial Optimization*** 2025
PhD Thesis, Technische Universität Berlin, 2025.
- The Submodular Santa Claus Problem*** with E. Bamas and L. Rohwedder 2025
In: *Proceedings of SODA 2025*.
- Single Source Unsplittable Flows with arc-wise lower and upper Bounds*** 2022
with M. Skutella
In: *Mathematical Programming*, 195(1), 1–35. Extended abstract: *IPCO 2020*.
- Minimum-cost Integer Circulations in Given Homology Classes*** 2021
with I. Seidel and S. Weltge
In: *Proceedings of SODA 2021*.
- Diversity Maximization in Doubling Metrics*** 2018
with A. Cevallos and F. Eisenbrand
In: *Proceedings of ISAAC 2018*.

EVENTS AND TALKS

- Hausdorff Research Institute for Mathematics, Bonn, Germany** April 2026
Follow-Up Workshop of the Program *Discrete Optimization* (by invitation)
Talk: *Unsplittable Transshipments*
- CNRS Centre Paul Langevin, Aussois, France** Jan 2026
Workshop *Combinatorial Optimization* (by invitation)
- Institut d'Études Scientifiques de Cargèse (IESC), France** Sep 2025
Workshop *Combinatorial Optimization* (by invitation)
Talk: *Unsplittable Transshipments*
- Max Planck Institute for Informatics, Saarbrücken, Germany** Nov 2024
Talk: *The Submodular Santa Claus Problem*
- École Polytechnique Fédérale de Lausanne, Switzerland** Sep 2024
Summer School *Synergies of Combinatorics and TCS*
- CNRS Centre Paul Langevin, Aussois, France** Jan 2024
Workshop *Combinatorial Optimization* (by invitation)
- École Polytechnique Fédérale de Lausanne, Switzerland** Jul 2022
Summer School and Workshop *Modern Trends in Combinatorial Optimization* (workshop by invitation)
- FUTURIUM, Berlin, Germany** Jul 2022
German-wide event *The 7 Greatest Mathematical Adventures*
Public introductory talk: *P versus NP* with Prof. Dr. Martin Skutella
- Hausdorff Research Institute for Mathematics, Bonn, Germany** Dec 2021
Workshop *Parametrized Complexity and Discrete Optimization* as part of the Program *Discrete Optimization* (by invitation)

Hausdorff Research Institute for Mathematics, Bonn, Germany Workshop <i>Approximation and Relaxation</i> as part of the Program <i>Discrete Optimization</i> (by invitation) Talk: <i>Unsplittable Flows and Their Application in Machine Scheduling</i>	Nov 2021
Mathematisches Forschungszentrum Oberwolfach, Germany Workshop <i>Combinatorial Optimization</i> (by invitation) Talk: <i>Single Source Unsplittable Flows</i>	Nov 2021
Technical University Chemnitz, Germany (online) DMV (German Mathematical Society) Annual Meeting Talk: <i>Single Source Unsplittable Flows</i>	Sep 2020
Schloss Dagstuhl, Leibniz Center for Informatics, Germany Seminar <i>Scheduling</i> (by invitation)	Feb 2020
CNRS Centre Paul Langevin, Aussois, France Workshop <i>Combinatorial Optimization</i> (by invitation) Talk: <i>Single Source Unsplittable Flows</i>	Jan 2020
Technical University Munich, Germany Talk: <i>Single Source Unsplittable Flows</i>	Nov 2019
University of Michigan, Ann Arbor, USA IPCO Summer School	May 2019
École Polytechnique Fédérale de Lausanne, Switzerland Talk: <i>Single Source Unsplittable Flows</i>	Mar 2019
Simons Institute for the Theory of Computing, Berkeley, USA Reunion Workshop <i>Bridging Continuous and Discrete Optimization</i>	Dec 2018
Mathematisches Forschungszentrum Oberwolfach, Germany Workshop <i>Combinatorial Optimization</i> (by invitation)	Nov 2018
Simons Institute for the Theory of Computing, Berkeley, USA Program <i>Bridging Continuous and Discrete Optimization</i>	Aug – Dec 2017

TEACHING AT UNIVERSITY OF BREMEN, GERMANY

Seminar on Metaheuristics (Course Instructor) Bachelor's program in Business Informatics and CS, 9 participants	Spring 2026
Operations Research (Course Instructor with Prof. Dr. Nicole Megow) Bachelor's program in Business Informatics and CS, 80 participants	Fall 2025
Algorithmic and Discrete Mathematics (ADM) (TA) Bachelor's program in Mathematics, 5 participants	Spring 2025

TEACHING AT TECHNISCHE UNIVERSITÄT BERLIN, GERMANY

Introduction to Linear and Discrete Optimization (ADM I) (TA) Bachelor's program in Mathematics, 50 participants	Fall 2024
Stochastics for Computer Science (TA) Bachelor's program in Engineering Sciences, 600 participants	Spring 2024

Analysis I and Linear Algebra (TA) Bachelor's program in Engineering Sciences, 1000 participants	Fall 2022
Discrete Optimization (ADM II) (TA) Master's program in Mathematics, 20 participants	Spring 2022
Computer-Oriented Mathematics (TA and plenary tutorial) Bachelor's program in Mathematics, 250 participants	Spring 2019 – Fall 2020
Linear Algebra (TA and plenary tutorial) Bachelor's program in Engineering Sciences, 800 participants	Fall 2018

TEACHING AT EPFL, SWITZERLAND

Advanced Linear Algebra (TA) Bachelor's program in Mathematics, 80 participants	Spring 2018
Discrete Optimization (TA) Bachelor's program in Mathematics, 40 participants	Spring 2018

OTHER SKILLS

Programming

Python (working knowledge), C/C++ (basic knowledge)

Languages

German (native), English (fluent), French (fluent)